Example 1.23 Verify $A-(B\cup C)=(A-B)\cap(A-C)$ using Venn diagrams.

Solution

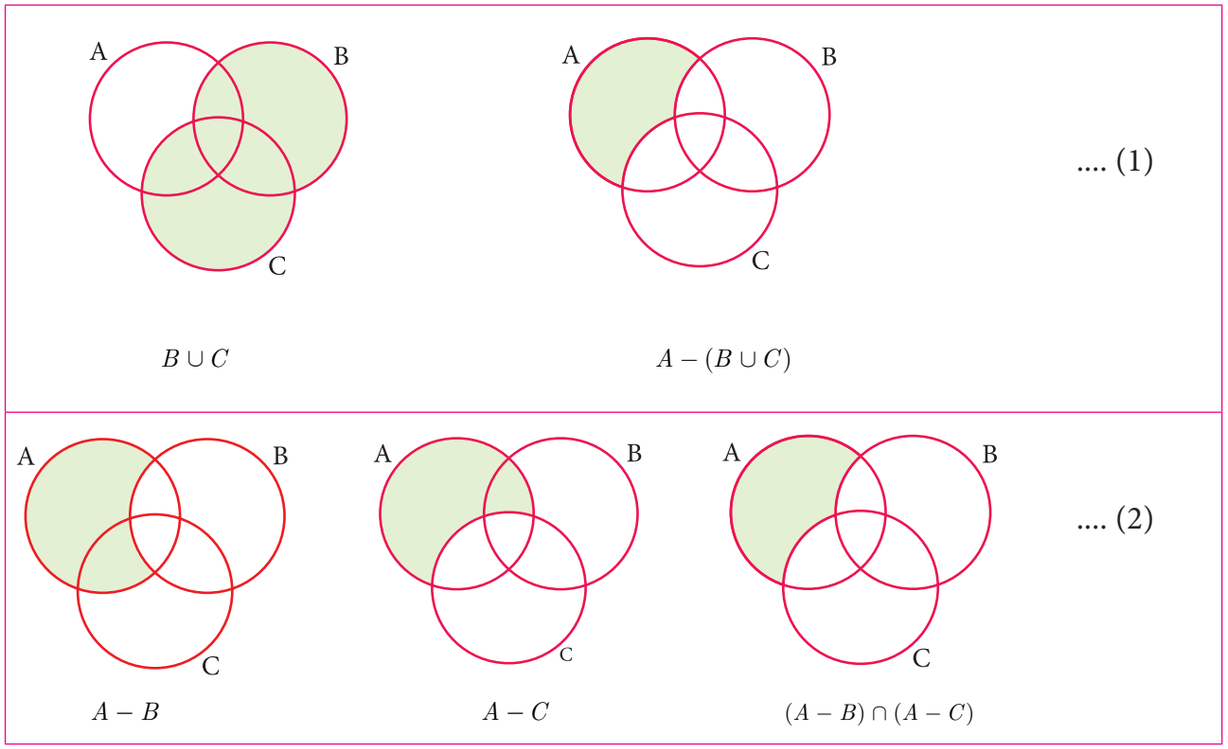


Fig.1.30

From (1) and (2), we get $A-(B\cup C)=(A-B)\cap(A-C)$. Hence it is verified.

Example 1.24 If $P=\{x : x\in \mathbb{W}$ and $0<x<10\}$, $Q=\{x : x=2n+1, n\in \mathbb{W}$ and $n<5\}$ and $R=\{2,3,5,7,11,13\}$, then verify $P-(Q\cap R)=(P-Q)\cup(P-R)$

Solution The roster form of sets P, Q and R are

$P=\{1,2,3,4,5,6,7,8,9\}$, $Q=\{1,3,5,7,9\}$

and $R=\{2,3,5,7,11,13\}$

First, we find $(Q\cap R)=\{3,5,7\}$

Then, $P-(Q\cap R)=\{1,2,4,6,8,9\}$ ... (1)

Next, $P-Q=\{2,4,6,8\}$ and

$P-R=\{1,4,6,8,9\}$

and so, $(P-Q)\cup(P-R)=\{1,2,4,6,8,9\}$ ... (2)

Finding the elements of set Q

Given, $x=2n+1$

$n=0\rightarrow x=2(0)+1=0+1=1$

$n=1\rightarrow x=2(1)+1=2+1=3$

$n=2\rightarrow x=2(2)+1=4+1=5$

$n=3\rightarrow x=2(3)+1=6+1=7$

$n=4\rightarrow x=2(4)+1=8+1=9$

Therefore, x takes values such as 1, 3, 5, 7 and 9.

Hence from (1) and (2), it is verified that $P-(Q\cap R)=(P-Q)\cup(P-R)$.

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